WEB TECHNOLOGIES LAB MIDTERM

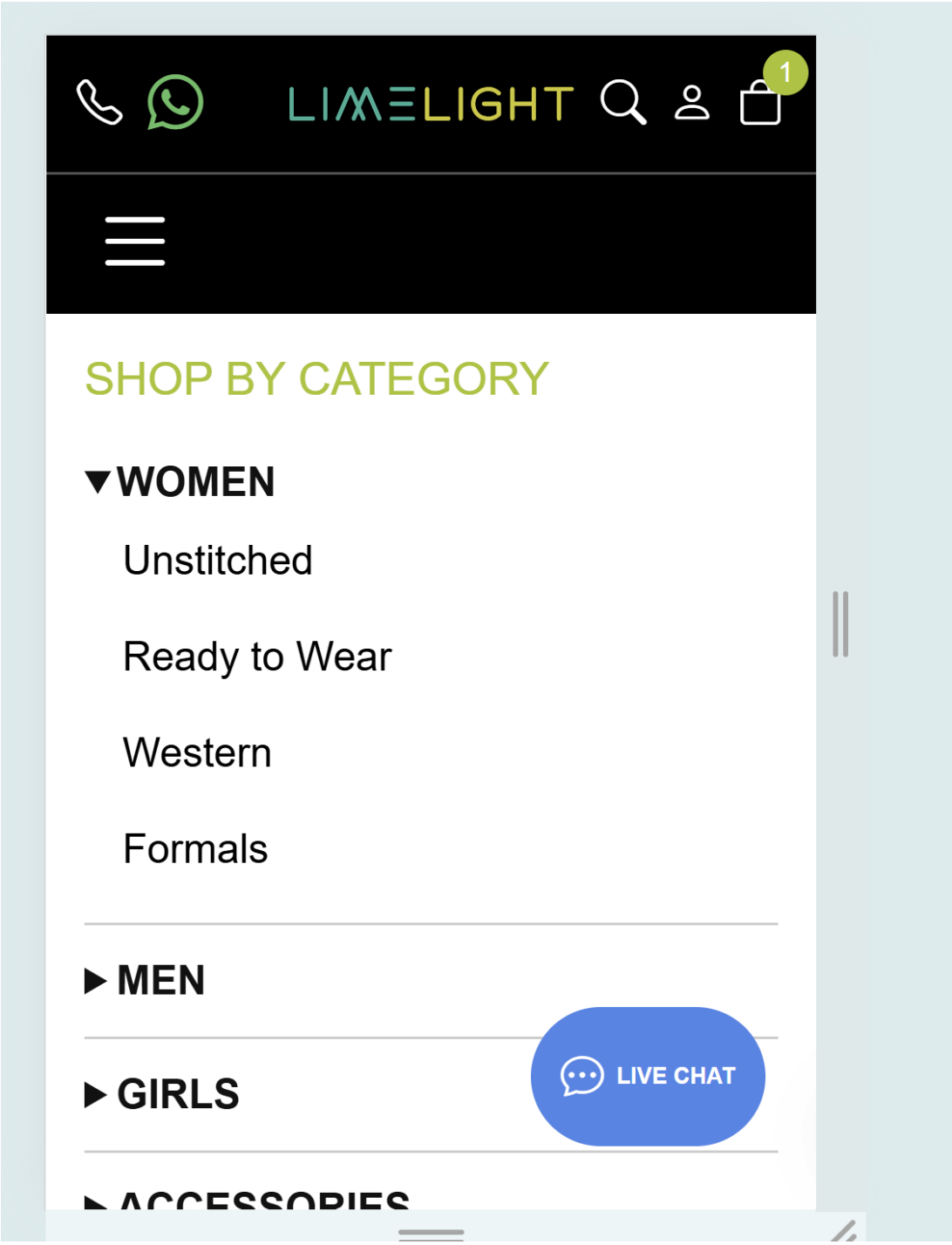
Completion Document

Student Name	Urwa Faisal
Registration No.	FA24-BCS-110
N	10
Breakpoint (BP)	610px
Accent Hue (H)	70 → hsl(70, 70%, 45%)
K	4

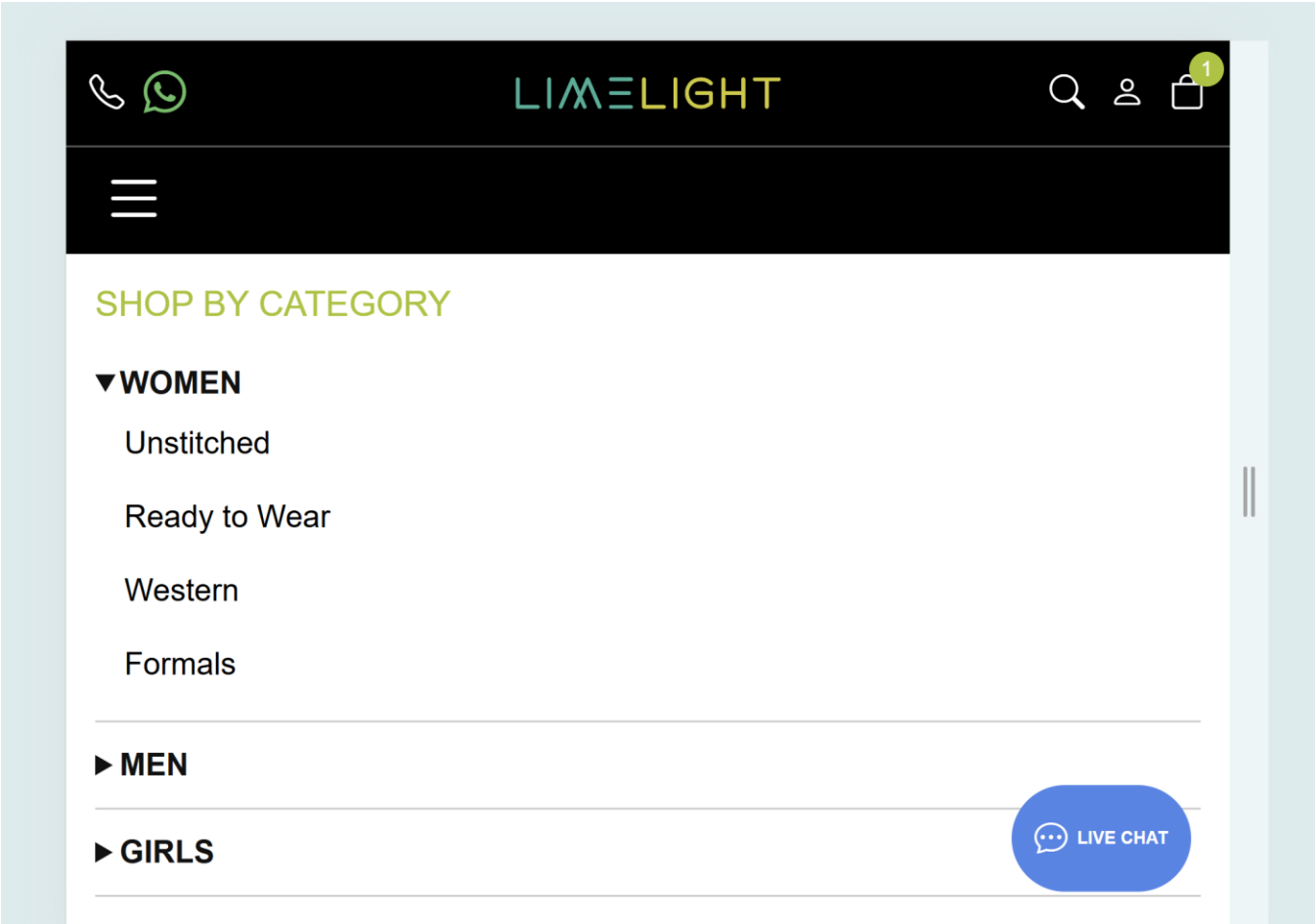
Project: Upgrade E-commerce Navigation, Checkout and Cart

Task 1 — Mega-menu / Shop by Category

Implemented responsive Shop by Category navigation. Below the breakpoint, category groups are presented as independent HTML details/summary accordions. At BP and wider, the desktop mega-menu uses four category groups and the required accent colour.



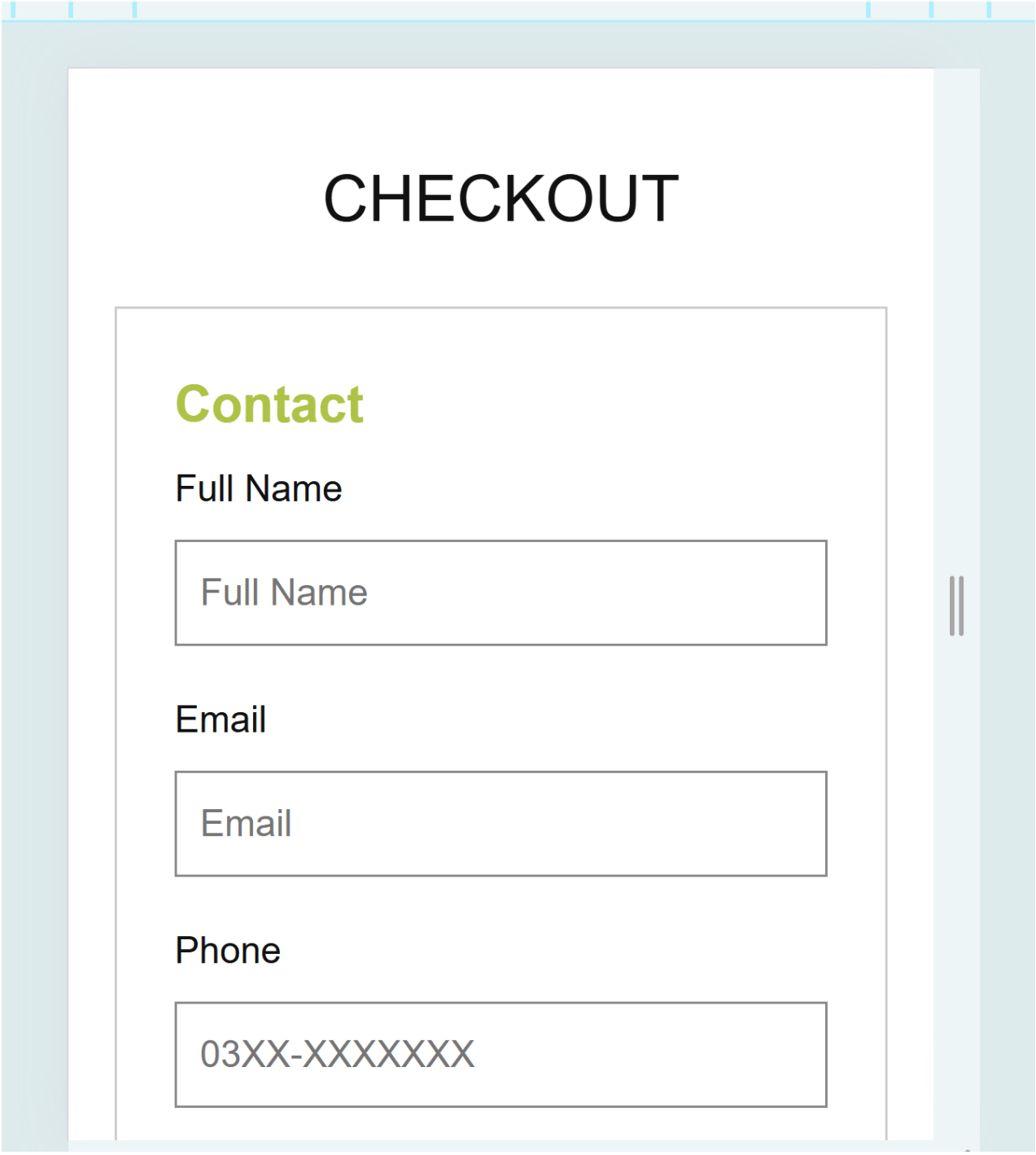
Task 1 — Mobile View (375px): Shop by Category accordion



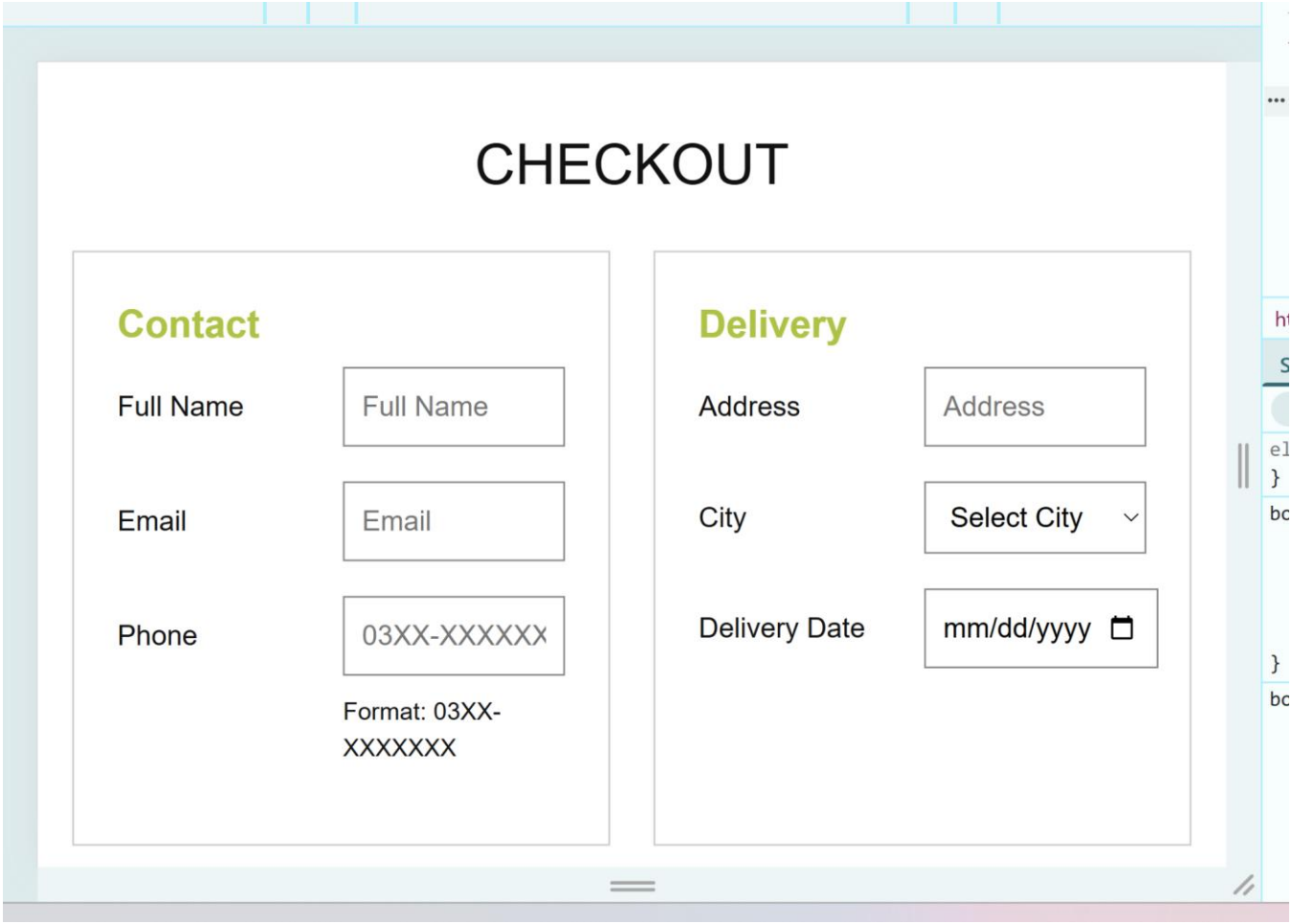
Task 1 — BP View (610px): responsive category navigation

Task 2 — Checkout Form

Created checkout.html using HTML5 and CSS validation only. The form contains Contact and Delivery fieldsets, associated labels, phone pattern validation, city selection and a restricted delivery date. The layout stacks below BP and becomes two columns from BP upward.



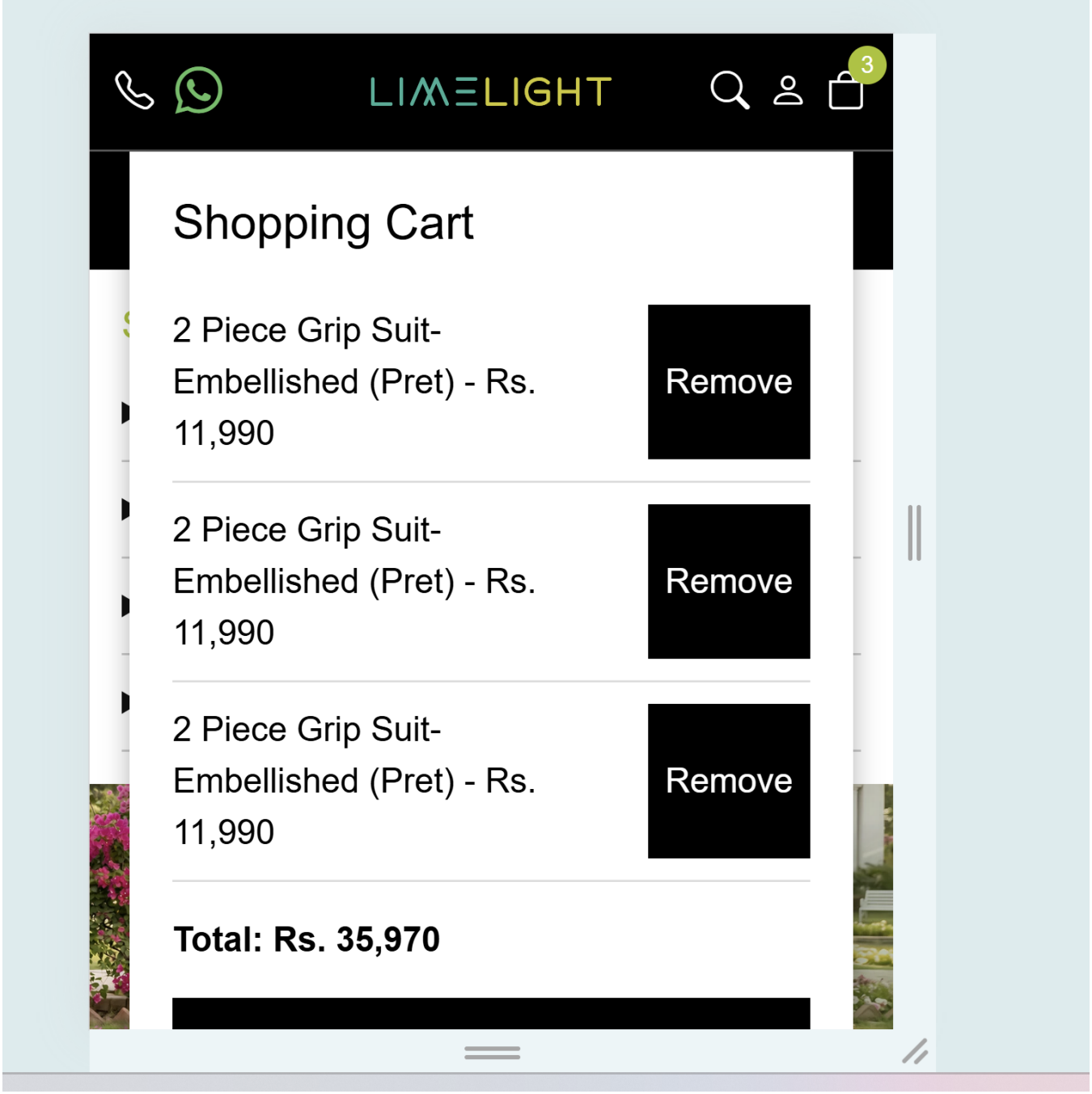
Task 2 — Mobile View (375px): stacked checkout form



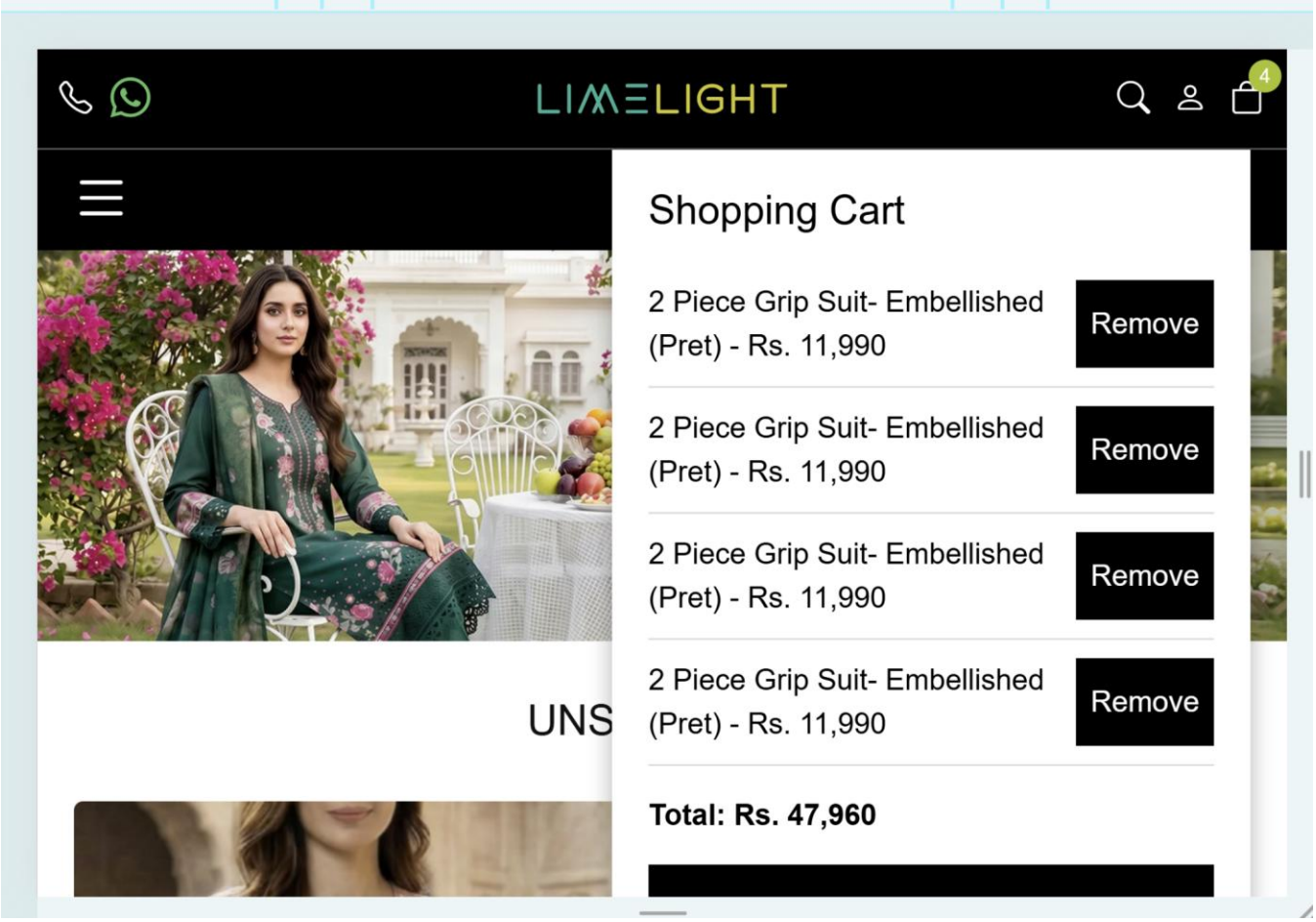
Task 2 — Desktop View (1280px): side-by-side Contact and Delivery fieldsets

Task 3 — JavaScript Cart Badge and Mini-cart

Added a plain-JavaScript cart. Products are stored in an array, the navbar badge updates with the item count, the mini-cart lists products with Remove buttons, and the total is calculated using reduce(). The same-product limit for N = 10 is 4.



Task 3 — Mobile View: mini-cart opened after adding products



Task 3 — Desktop View (1280px): cart badge, mini-cart, Remove buttons and total

Code Added / Changed

Task 1 — HTML Added

```
<li class="nav-item shop-menu">
    <a class="nav-link shop-link" href="#">SHOP BY CATEGORY</a>

    <div class="mega-menu">
        <div class="mega-column">
            <h3>WOMEN</h3>
            <ul>
                <li><a href="#">Unstitched</a></li>
                <li><a href="#">Ready to Wear</a></li>
                <li><a href="#">Western</a></li>
                <li><a href="#">Formals</a></li>
            </ul>
        </div>

        <div class="mega-column">
            <h3>MEN</h3>
            <ul>
                <li><a href="#">Kurtas</a></li>
                <li><a href="#">Shirts</a></li>
                <li><a href="#">Trousers</a></li>
                <li><a href="#">Waistcoats</a></li>
            </ul>
        </div>

        <div class="mega-column">
            <h3>GIRLS</h3>
            <ul>
                <li><a href="#">Dresses</a></li>
                <li><a href="#">Kurtas</a></li>
                <li><a href="#">Tops</a></li>
                <li><a href="#">Bottoms</a></li>
            </ul>
        </div>

        <div class="mega-column">
            <h3>ACCESSORIES</h3>
            <ul>
                <li><a href="#">Bags</a></li>
                <li><a href="#">Jewellery</a></li>
                <li><a href="#">Scarves</a></li>
                <li><a href="#">Footwear</a></li>
            </ul>
        </div>
    </div>
</li>

<div class="mobile-categories">
    <h2>SHOP BY CATEGORY</h2>
    <details>
        <summary>WOMEN</summary>
        <a href="#">Unstitched</a>
        <a href="#">Ready to Wear</a>
        <a href="#">Western</a>
        <a href="#">Formals</a>
    </details>
    <details>
        <summary>MEN</summary>
        <a href="#">Kurtas</a>
        <a href="#">Shirts</a>
        <a href="#">Trousers</a>
        <a href="#">Waistcoats</a>
    </details>
    <details>
        <summary>GIRLS</summary>
        <a href="#">Dresses</a>
        <a href="#">Kurtas</a>
        <a href="#">Tops</a>
        <a href="#">Bottoms</a>
    </details>
    <details>
        <summary>ACCESSORIES</summary>
        <a href="#">Bags</a>
        <a href="#">Jewellery</a>
        <a href="#">Scarves</a>
        <a href="#">Footwear</a>
```

```
    </details>
</div>
```

Task 1 — CSS Added

```
.shop-menu { position: static; }

.mega-menu {
    display: none;
    position: absolute;
    left: 0;
    top: 100%;
    width: 100%;
    background: white;
    padding: 30px 50px;
    grid-template-columns: repeat(4, 1fr);
    gap: 30px;
    z-index: 1000;
    box-shadow: 0 5px 10px #999;
}

.shop-menu:hover .mega-menu,
.shop-menu:focus-within .mega-menu {
    display: grid;
}

.mega-column h3 {
    color: hsl(70, 70%, 45%);
}

.mobile-categories { display: none; }

@media (max-width: 609px) {
    .shop-menu { display: none; }
    .mobile-categories { display: block; }
}
```

Task 2 — checkout.html Added

```
<main class="checkout-page">
    <h1>CHECKOUT</h1>

    <form class="checkout-form">
        <fieldset>
            <legend>Contact</legend>

            <div class="form-row">
                <label for="fullname">Full Name</label>
                <input type="text" id="fullname" name="fullname"
                       placeholder="Full Name" required>
            </div>

            <div class="form-row">
                <label for="email">Email</label>
                <input type="email" id="email" name="email"
                       placeholder="Email" required>
            </div>

            <div class="form-row">
                <label for="phone">Phone</label>
                <input type="tel" id="phone" name="phone"
                       placeholder="03XX-XXXXXXX"
                       pattern="03[0-9]{2}-[0-9]{7}" required>
                <small>Format: 03XX-XXXXXXX</small>
            </div>
        </fieldset>

        <fieldset>
            <legend>Delivery</legend>

            <div class="form-row">
                <label for="address">Address</label>
                <input type="text" id="address" name="address"
                       placeholder="Address" required>
            </div>

            <div class="form-row">
                <label for="city">City</label>
                <select id="city" name="city" required>
                    <option value="">Select City</option>
                    <option value="Lahore">Lahore</option>
                    <option value="Islamabad">Islamabad</option>
                    <option value="Karachi">Karachi</option>
                    <option value="Faisalabad">Faisalabad</option>
                </select>
            </div>

            <div class="form-row">
                <label for="date">Delivery Date</label>
                <input type="date" id="date" name="date"
                       min="2026-10-29"
                       max="2026-11-08" required>
            </div>
        </fieldset>

        <button type="submit" class="checkout-button">PLACE ORDER</button>
    </form>
</main>
```

Task 2 — CSS Added

```
.checkout-page {
    max-width: 1200px;
    margin: auto;
    padding: 40px 20px;
}

.checkout-form {
    display: grid;
    grid-template-columns: 1fr 1fr;
    gap: 25px;
}

.checkout-form fieldset {
    border: 1px solid #ccc;
    padding: 25px;
}
```

```
.checkout-form legend {
    font-size: 22px;
    font-weight: bold;
    color: hsl(70, 70%, 45%);
}

.form-row {
    display: grid;
    grid-template-columns: 120px 1fr;
    gap: 10px;
    margin-bottom: 20px;
    align-items: center;
}

.form-row input,
.form-row select {
    width: 100%;
    padding: 10px;
    border: 1px solid #888;
}

.form-row small { grid-column: 2; }

.checkout-button {
    grid-column: 1 / -1;
    width: 100%;
    padding: 15px;
    border: none;
    background: black;
    color: white;
}

.form-row input:user-invalid,
.form-row select:user-invalid {
    border: 2px solid red;
}

@media (max-width: 609px) {
    .checkout-form { grid-template-columns: 1fr; }
    .form-row { grid-template-columns: 1fr; }
    .form-row small { grid-column: 1; }
    .checkout-button { grid-column: 1; }
}
```

Task 3 — HTML Added / Changed

```
<div class="cart-icon" id="cartIcon">
    <i class="bi bi-bag"></i>
    <span id="cartCount">0</span>
</div>

<div class="mini-cart" id="miniCart">
    <h2>Shopping Cart</h2>
    <div id="cartItems">
        <p>Your cart is empty.</p>
    </div>
    <div class="cart-total">
        Total: Rs. <span id="cartTotal">0</span>
    </div>
    <a href="checkout.html" class="checkout-btn">CHECKOUT</a>
</div>

<!-- Product card additions -->
<button type="button" class="mini-bag add-to-cart"
        onclick="event.preventDefault(); event.stopPropagation();">
    <i class="bi bi-bag-plus"></i>
</button>
<p class="product-name">2 Piece Grip Suit- Embellished (Pret)</p>
<h3 class="product-price">Rs. 11,990</h3>

<script src="script.js"></script>
```

Task 3 — CSS Added

```
.cart-icon {
    position: relative;
    cursor: pointer;
}

#cartCount {
    position: absolute;
    top: -10px;
    right: -10px;
    background: hsl(70, 70%, 45%);
    color: white;
    border-radius: 50%;
    padding: 2px 6px;
    font-size: 11px;
}

.mini-cart {
    display: none;
    position: absolute;
    right: 20px;
    top: 70px;
    width: 350px;
    background: white;
    color: black;
    padding: 20px;
    z-index: 2000;
    box-shadow: 0 4px 12px #777;
}

.mini-cart.show-cart { display: block; }

.cart-item {
    display: flex;
    justify-content: space-between;
    gap: 10px;
    padding: 10px 0;
    border-bottom: 1px solid #ddd;
}

.cart-item button {
    border: none;
    background: black;
    color: white;
    padding: 5px 8px;
}

.cart-total {
    font-weight: bold;
    margin: 20px 0;
}
```

```
.checkout-btn {
    display: block;
    background: black;
    color: white;
    text-align: center;
    text-decoration: none;
    padding: 12px;
}

.mini-bag.limit-reached {
    width: auto;
    padding: 0 10px;
    border-radius: 18px;
    font-size: 11px;
    background: #ddd;
    cursor: not-allowed;
}

@media (max-width: 609px) {
    .mini-cart {
        width: 90%;
        right: 5%;
    }
}
```

Task 3 — script.js

```
let cart = [];

const cartIcon = document.getElementById("cartIcon");
const miniCart = document.getElementById("miniCart");
const cartCount = document.getElementById("cartCount");
const cartItems = document.getElementById("cartItems");
const cartTotal = document.getElementById("cartTotal");
const buttons = document.querySelectorAll(".add-to-cart");

cartIcon.addEventListener("click", function () {
    miniCart.classList.toggle("show-cart");
});

buttons.forEach(function (button) {
    button.addEventListener("click", function (event) {
        event.preventDefault();
        event.stopPropagation();

        const card = button.closest(".product-card");
        const name = card.querySelector(".product-name").textContent;
        const priceText = card.querySelector(".product-price").textContent;
        const price = Number(priceText.replace(/[^0-9]/g, ""));

        const sameProducts = cart.filter(function (item) {
            return item.name === name;
        });

        if (sameProducts.length >= 4) return;

        cart.push({ name: name, price: price });
        displayCart();
        updateButtons();
    });
});

function displayCart() {
    cartItems.innerHTML = "";

    cart.forEach(function (item, index) {
        const div = document.createElement("div");
        div.classList.add("cart-item");
        div.innerHTML =
            `<span>${item.name} - Rs. ${item.price.toLocaleString()}</span>
             <button onclick="removeItem(${index})">Remove</button>`;
        cartItems.appendChild(div);
    });

    cartCount.textContent = cart.length;

    const total = cart.reduce(function (sum, item) {
        return sum + item.price;
    }, 0);
```

```
    cartTotal.textContent = total.toLocaleString();

    if (cart.length === 0) {
        cartItems.innerHTML = "<p>Your cart is empty.</p>";
    }
}

function updateButtons() {
    buttons.forEach(function (button) {
        const card = button.closest(".product-card");
        const name = card.querySelector(".product-name").textContent;

        const count = cart.filter(function (item) {
            return item.name === name;
        }).length;

        if (count >= 4) {
            button.disabled = true;
            button.innerHTML = "Limit reached";
            button.classList.add("limit-reached");
        } else {
            button.disabled = false;
            button.innerHTML = '<i class="bi bi-bag-plus"></i>';
            button.classList.remove("limit-reached");
        }
    });
}

function removeItem(index) {
    cart.splice(index, 1);
    displayCart();
    updateButtons();
}
```

Files Changed

• index.html

• style.css

• checkout.html

• script.js

AI Tool Used

ChatGPT

Short Task Summary

Task 1: Added responsive Shop by Category navigation using a desktop mega-menu and mobile details/summary accordion. The breakpoint is based on BP = 610px and the accent colour uses H = 70.

Task 2: Added a responsive checkout form with Contact and Delivery fieldsets. HTML5 validation is used for required fields, phone format and delivery date restrictions without custom JavaScript.

Task 3: Added a plain-JavaScript cart with an item-count badge, toggleable mini-cart, Remove actions and a reduce()-based total. The same-product maximum is 4 for N = 10.